INCH-POUND
A-A-59931
29 July 2013

# COMMERCIAL ITEM DESCRIPTION

CURRENT-VOLTAGE STANDARD

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for PD13WRGUEEA14, PD10WRGBGBEC29 and PD04WRLEEC03 for all federal agencies.

## 1. SCOPE.

This commercial item description (CID) describes a Current-Voltage Standard, hereinafter referred to as an Electronic Calibrator. The Electronic Calibrator provides accurate reference values of voltage, resistance and current for the calibration of test measurement and diagnostic equipment for the development, repair and maintenance of navigational aids, radio frequency communications equipment and other measurement equipment. This CID lists the minimum performance, design, functional, environmental, reliability requirements and quality assurance provisions for the Electronic Calibrator.

## 2. CLASSIFICATION.

The Electronic Calibrator covered by this CID will be commercially available equipment that has factory installed options and/or may be modified to the extent necessary to meet the following description. The test set shall be Class 3, in accordance with MIL-PRF-28800, except as specified herein.

## 3. SALIENT CHARACTERISTICS.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: WR-ALC/404 SCMS/GUEEA, Robins AFB GA 31098-1670. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil .

AMSC N/A FSC 6625

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3.1 Item Description. The Electronic Calibrator provides five electrical functions; alternating voltage, alternating current, direct voltage, direct current and resistance values. The alternating voltage and current functions have a frequency span from 10 Hz to 1 MHz, the extended bandwidth options extends the frequency response to 30 MHz. This electrical calibrator is used in the development, repair, calibration and maintenance of test measurement and diagnostic equipment in support of avionics, microwave, navigational aids, r*adar,* satellite and other communications equipment.

3.2 Electrical Calibrator. The electronic calibrator shall be capable of providing DC voltages with confidence levels as shown in table 1 Voltage Confidence & Resolution.

Table 1 DC Voltage Confidence & Resolution

| Range | Resolution | Absolute Uncertainty ± 5 °C from calibration temperature [1] | | | | Relative Uncertainty ± 1 °C | |
|---|---|---|---|---|---|---|---|
| | | 24 Hours | 90 Days | 180 Days | 1 Year | 24 Hours | 90 Days |
| | | ± (ppm output + µV) | | | | | |
| 99 % Confidence Level | | | | | | | |
| 220 mV | 10 nV | 5 + 0.5 | 7 + 0.5 | 8 + 0.5 | 9 + 0.5 | 2 + 0.4 | 2.5 + 0.4 |
| 2.2 V | 100 nV | 3.5 + 0.8 | 4 + 0.8 | 4.5 + 0.8 | 6 + 0.8 | 2 + 0.8 | 2.5 + 0.8 |
| 11 V | 1 µV | 2.5 + 3 | 3 + 3 | 3.5 + 3 | 4 + 3 | 1 + 3 | 1.5 + 3 |
| 22 V | 1 µV | 2.5 + 5 | 3 + 5 | 3.5 + 5 | 4 + 5 | 1 + 5 | 1.5 + 5 |
| 220 V | 10 µV | 3.5 + 50 | 4 + 50 | 5 + 50 | 6 + 50 | 2 + 50 | 2.5 + 50 |
| 1100 V | 100 µV | 5 + 500 | 6 + 500 | 7 + 500 | 8 + 500 | 2.5 + 400 | 3 + 400 |
| 95 % Confidence Level | | | | | | | |
| 220 mV | 10 nV | 4 + 0.4 | 6 + 0.4 | 6.5 + 0.4 | 7.5 + 0.4 | 1.6 + 0.4 | 2 + 0.4 |
| 2.2 V | 100 nV | 3 + 0.7 | 3.5 + 0.7 | 4 + 0.7 | 5 + 0.7 | 1.6 + 0.7 | 2 + 0.7 |
| 11 V | 1 µV | 2 + 2.5 | 2.5 + 2.5 | 3 + 2.5 | 3.5 + 2.5 | 0.8 + 2.5 | 1.2 + 2.5 |
| 22 V | 1 µV | 2 + 4 | 2.5 + 4 | 3 + 4 | 3.5 + 4 | 0.8 + 4 | 1.2 + 4 |
| 220 V | 10 µV | 3 + 40 | 3.5 + 40 | 4 + 40 | 5 + 40 | 1.6 + 40 | 2 + 40 |
| 1100 V | 100 µV | 4 + 400 | 4.5 + 400 | 6 + 400 | 6.5 + 400 | 2 + 400 | 2.4 + 400 |

Notes:
DC Zeros calibration required every 30 days.
1. For fields strengths >1 V/m but ≤3 V/m, in the band of 80 MHz to 1 GHz, add 0.01 % of range.

3.3 Frequency Range. The electronic calibrator shall have a minimum effective frequency range from 10 Hz to 1 MHz analog mode (AC coupled and with temperature and frequency compensation enabled) as shown on table 2.



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Table 2 AC Voltage Confidence & Resolution

| Range | Resolution | Frequency (Hz) | Absolute Uncertainty ± 5 °C from calibration temperature | | | | Relative Uncertainty ± 1 °C | |
|---|---|---|---|---|---|---|---|---|
| | | | 24 Hours | 90 Days | 180 Days | 1 Year | 24 Hours | 90 Days |
| | | | ± (ppm output + µV) | | | | | |
| 2.2 mV | 1 nV | 10 - 20 | 500 + 5 | 550 + 5 | 600 + 5 | 600 + 5 | 500 + 5 | 550 + 5 |
| | | 20 - 40 | 200 + 5 | 220 + 5 | 230 + 5 | 240 + 5 | 200 + 5 | 220 + 5 |
| | | 40 - 20 k | 100 + 5 | 110 + 5 | 120 + 5 | 120 + 5 | 60 + 5 | 65 + 5 |
| | | 20 k - 50 k | 340 + 5 | 370 + 5 | 390 + 5 | 410 + 5 | 100 + 5 | 110 + 5 |
| | | 50 k - 100 k | 800 + 8 | 900 + 8 | 950 + 8 | 950 + 8 | 220 + 8 | 240 + 8 |
| | | 100 k - 300 k | 1100 + 15 | 1200 + 15 | 1300 + 15 | 1300 + 15 | 400 + 15 | 440 + 15 |
| | | 300 k - 500 k | 1500 + 30 | 1700 + 30 | 1700 + 30 | 1800 + 30 | 1000 + 30 | 1100 + 30 |
| | | 500 k - 1 M | 4000 + 40 | 4400 + 40 | 4700 + 40 | 4800 + 40 | 400 + 30 | 4400 + 30 |
| 22 mV | 10 nV | 10 - 20 | 500 + 6 | 550 + 6 | 600 + 6 | 600 + 6 | 500 + 6 | 550 + 6 |
| | | 20 - 40 | 200 + 6 | 220 + 6 | 230 + 6 | 240 + 6 | 200 + 6 | 220 + 6 |
| | | 40 - 20 k | 100 + 6 | 110 + 6 | 120 + 6 | 120 + 6 | 60 + 6 | 65 + 6 |
| | | 20 k - 50 k | 340 + 6 | 370 + 6 | 390 + 6 | 410 + 6 | 100 + 6 | 110 + 6 |
| | | 50 k - 100 k | 800 + 8 | 900 + 8 | 950 + 8 | 950 + 8 | 220 + 8 | 240 + 8 |
| | | 100 k - 300 k | 1100 + 15 | 1200 + 15 | 1300 + 15 | 1300 + 15 | 400 + 15 | 440 + 15 |
| | | 300 k - 500 k | 1500 + 30 | 1700 + 30 | 1700 + 30 | 1800 + 30 | 1000 + 30 | 1100 + 30 |
| | | 500 k - 1 M | 4000 + 40 | 4400 + 40 | 4700 + 40 | 4800 + 40 | 4000 + 30 | 4400 + 30 |
| 220 mV | 100 nV | 10 - 20 | 500 + 16 | 550 + 16 | 600 + 16 | 600 + 16 | 500 + 16 | 550 + 16 |
| | | 20 - 40 | 200 + 10 | 220 + 10 | 230 + 10 | 240 + 10 | 200 + 10 | 220 + 10 |
| | | 40 - 20 k | 95 + 10 | 100 + 10 | 110 + 10 | 110 + 10 | 60 + 10 | 65 + 10 |
| | | 20 k - 50 k | 300 + 10 | 330 + 10 | 350 + 10 | 360 + 10 | 100 + 10 | 110 + 10 |
| | | 50 k - 100 k | 750 + 30 | 800 + 30 | 850 + 30 | 900 + 30 | 220 + 30 | 240 + 30 |
| | | 100 k - 300 k | 940 + 30 | 1000 + 30 | 1100 + 30 | 1100 + 30 | 400 + 30 | 440 + 30 |
| | | 300 k - 500 k | 1500 + 40 | 1700 + 40 | 1700 + 40 | 1800 + 40 | 1000 + 40 | 1100 + 40 |
| | | 500 k - 1 M | 3000 + 100 | 3300 + 100 | 3500 + 100 | 3600 + 100 | 3000 + 100 | 3300 + 100 |
| 2.2 V | 1 µV | 10 - 20 | 500 + 100 | 550 + 100 | 600 + 100 | 600 + 100 | 500 + 100 | 550 + 100 |
| | | 20 - 40 | 150 + 30 | 170 + 30 | 170 + 30 | 180 + 30 | 150 + 30 | 170 + 30 |
| | | 40 - 20 k | 70 + 7 | 75 + 7 | 80 + 7 | 85 + 7 | 40 + 7 | 45 + 7 |
| | | 20 k - 50 k | 120 + 20 | 130 + 20 | 140 + 20 | 140 + 20 | 100 + 20 | 110 + 20 |
| | | 50 k - 100 k | 230 + 80 | 250 + 80 | 270 + 80 | 280 + 80 | 200 + 80 | 220 + 80 |
| | | 100 k - 300 k | 400 + 150 | 440 + 150 | 470 + 150 | 480 + 150 | 400 + 150 | 440 + 150 |
| | | 300 k - 500 k | 1000 + 400 | 1100 + 400 | 1200 + 400 | 1200 + 400 | 1000 + 400 | 1100 + 400 |
| | | 500 k - 1 M | 2000 + 1000 | 2200 + 1000 | 2300 + 1000 | 2400 + 1000 | 2000 + 1000 | 2200 + 1000 |
| 22 V | 10 µV | 10 - 20 | 500 + 1000 | 550 + 1000 | 600 + 1000 | 600 + 1000 | 500 + 1000 | 550 + 1000 |
| | | 20 - 40 | 150 + 300 | 170 + 300 | 170 + 300 | 180 + 300 | 150 + 300 | 170 + 300 |
| | | 40 - 20 k | 70 + 70 | 75 + 70 | 80 + 70 | 85 + 70 | 40 + 70 | 45 + 70 |
| | | 20 k - 50 k | 120 + 200 | 130 + 200 | 140 + 200 | 140 + 200 | 100 + 200 | 110 + 200 |
| | | 50 k - 100 k | 230 + 400 | 250 + 400 | 270 + 400 | 280 + 400 | 200 + 400 | 220 + 400 |
| | | 100 k - 300 k | 500 + 1700 | 550 + 1700 | 550 + 1700 | 600 + 1700 | 500 + 1700 | 550 + 1700 |
| | | 300 k - 500 k | 1200 + 5000 | 1300 + 5000 | 1300 + 5000 | 1400 + 5000 | 1200 + 5000 | 1300 + 5000 |
| | | 500 k - 1 M | 2600 + 9000 | 2800 + 9000 | 2900 + 9000 | 3000 + 9000 | 2600 + 9000 | 2800 + 9000 |
| | | | ± (ppm output + mV) | | | | | |
| 220 V[2] | 100 µV | 10 - 20 | 500 + 10 | 550 + 10 | 600 + 10 | 600 + 10 | 500 + 10 | 550 + 10 |
| | | 20 - 40 | 150 + 3 | 170 + 3 | 170 + 3 | 180 + 3 | 150 + 3 | 170 + 3 |
| | | 40 - 20 k | 75 + 1 | 80 + 1 | 85 + 1 | 90 + 1 | 45 + 1 | 50 + 1 |
| | | 20 k - 50 k | 200 + 4 | 220 + 4 | 240 + 4 | 250 + 4 | 100 + 1 | 110 + 1 |
| | | 50 k - 100 k | 500 + 10 | 550 + 10 | 600 + 10 | 600 + 10 | 300 + 10 | 330 + 10 |
| | | 100 k - 300 k | 1500 + 110 | 1500 + 110 | 1600 + 110 | 1600 + 110 | 1500 + 110 | 1500 + 100 |
| | | 300 k - 500 k | 5000 + 110 | 5200 + 110 | 5300 + 110 | 5400 + 110 | 5000 + 110 | 5200 + 110 |
| | | 500 k - 1 M | 12,000 + 220 | 12,500 + 220 | 12,500 + 220 | 13,000 + 220 | 12,000 + 220 | 12,000 + 220 |
| 1100 V[1] | 1 mV | 15 - 50 | 400 + 20 | 420 + 20 | 440 + 20 | 460 + 20 | 400 + 20 | 420 + 20 |
| | | 50 - 1 k | 75 + 4 | 80 + 4 | 85 + 4 | 90 + 4 | 50 + 4 | 55 + 4 |

3.4 Voltage Reading Stability. The output voltage, within ±5 degrees C of the calibration temperature, shall be within ± (.001% of setting + .0001% of range + 1 µV) for all ranges



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3.5 Stability Aging Accuracy. The electronic calibrator accuracy variance shall be less than ±1% of programmed value within the temperature range of 32° to 122°F (0°C to 50°C) after tests in section 4.5.5.1.1 has been performed for class 3 equipment classification (Per table 11 on page 41). The electronic calibrator stability shall be within the values listed in tables 1 thru 3 shown hereafter.

Table 3 AC Current Confidence and Variation.

| Range | Resolution | Frequency (Hz) | Absolute Uncertainty ± 5 °C from calibration temperature [1] | | | | Relative Uncertainty ± 1 °C | |
|---|---|---|---|---|---|---|---|---|
| | | | 24 Hours | 90 Days | 180 Days | 1 Year | 24 Hours | 90 Days |
| | | | ± (ppm output + nA) | | | | | |
| 220 µA | 1 nA | 10 - 20 | 260 + 20 | 280 + 20 | 290 + 20 | 300 + 20 | 260 + 20 | 280 + 20 |
| | | 20 - 40 | 170 + 12 | 180 + 12 | 190 + 12 | 200 + 12 | 130 + 12 | 150 + 12 |
| | | 40 - 1 k | 120 + 10 | 130 + 10 | 135 + 10 | 140 + 10 | 100 + 10 | 110 + 10 |
| | | 1k - 5 k | 300 + 15 | 320 + 15 | 340 + 15 | 350 + 15 | 250 + 15 | 280 + 15 |
| | | 5k - 10 k | 1000 + 80 | 1100 + 80 | 1200 + 80 | 1300 + 80 | 900 + 80 | 1000 + 80 |
| 2.2 mA | 10 nA | 10 - 20 | 260 + 50 | 280 + 50 | 290 + 50 | 300 + 50 | 260 + 50 | 280 + 50 |
| | | 20 - 40 | 170 + 40 | 180 + 40 | 190 + 40 | 200 + 40 | 130 + 40 | 150 + 40 |
| | | 40 - 1 k | 120 + 40 | 130 + 40 | 135 + 40 | 140 + 40 | 100 + 40 | 110 + 40 |
| | | 1k - 5 k | 210 + 130 | 220 + 130 | 230 + 130 | 240 + 130 | 190 + 130 | 220 + 130 |
| | | 5k - 10 k | 1000 + 800 | 1100 + 800 | 1200 + 800 | 1300 + 800 | 900 + 800 | 1000 + 800 |
| 22 mA | 100 nA | 10 - 20 | 260 + 500 | 280 + 500 | 290 + 500 | 300 + 500 | 260 + 500 | 280 + 500 |
| | | 20 - 40 | 170 + 400 | 180 + 400 | 190 + 400 | 200 + 400 | 130 + 400 | 150 + 400 |
| | | 40 - 1 k | 120 + 400 | 130 + 400 | 135 + 400 | 140 + 400 | 100 + 400 | 110 + 400 |
| | | 1k - 5 k | 210 + 700 | 220 + 700 | 230 + 700 | 240 + 700 | 190 + 700 | 220 + 700 |
| | | 5k - 10 k | 1000 + 6000 | 1100 + 6000 | 1200 + 6000 | 1300 + 6000 | 900 + 6000 | 1000 + 6000 |
| | | | ± (ppm output + µA) | | | | | |
| 220 mA | 1 µA | 10 - 20 | 260 + 5 | 280 + 5 | 290 + 5 | 300 + 5 | 260 + 5 | 280 + 5 |
| | | 20 - 40 | 170 + 4 | 180 + 4 | 190 + 4 | 200 + 4 | 130 + 4 | 150 + 4 |
| | | 40 - 1 k | 120 + 3 | 130 + 3 | 135 + 3 | 140 + 3 | 100 + 3 | 110 + 3 |
| | | 1k - 5 k | 210 + 4 | 220 + 4 | 230 + 4 | 240 + 4 | 190 + 4 | 220 + 4 |
| | | 5k - 10 k | 1000 + 12 | 1100 + 12 | 1200 + 12 | 1300 + 12 | 900 + 12 | 1000 + 12 |
| 2.2 A | 10 µA | 20 - 1 k | 290 + 40 | 300 + 40 | 310 + 40 | 320 + 40 | 260 + 40 | 280 + 40 |
| | | 1 k - 5 k | 440 + 100 | 460 + 100 | 480 + 100 | 500 + 100 | 420 + 100 | 440 + 100 |
| | | 5 k - 10 k | 6000 + 200 | 7000 + 200 | 7500 + 200 | 8000 + 200 | 6000 + 200 | 7000 + 200 |
| 5725A Amplifier: | | | | | | | | |
| 11 A | 100 µA | 40 - 1 k | 370 + 170 | 400 + 170 | 440 + 170 | 460 + 170 | 300 + 170 | 330 + 170 |
| | | 1 k - 5 k | 800 + 380 | 850 + 380 | 900 + 380 | 950 + 380 | 700 + 380 | 800 + 380 |
| | | 5 k - 10 k | 3000 + 750 | 3300 + 750 | 3500 + 750 | 3600 + 750 | 2800 + 750 | 3200 + 750 |

Note:

Maximum output from the calibrator's terminals is 2.2 A. Uncertainty specifications for 220 µA and 2.2 mA ranges are increased by a factor of 1.3 plus 2 µA when supplied through 5725A terminals. For the 5720A 220 µA range, 1 kHz through 5 kHz and 5 kHz through 10 kHz, when the output is coming from the AUX current terminal, use the 5700A Absolute Uncertainty Specifications. Specifications are otherwise identical for all output locations.

1. For fields strengths >0.4 V/m but ≤3 V/m, in the band of 80 MHz to 1 GHz, add 1 % of range.

3.6 AC Voltage Measurement Uncertainty. The electronic calibrator shall meet or exceed the following AC measurement uncertainty specifications displayed on table 3 and Absolute Uncertainty Specifications.

3.7 Display Information. The electronic calibrator shall provide voltage magnitude reading, frequency, range of measurement, average value and reference value of comparison standard on the display. Menus for built in diagnostics shall be available on demand by the user and performed automatically at "Power On" or restart/reset condition.



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3.8 Triggering. The electronic calibrator shall have triggering capability to enable measurement functions through the GPIB or Ethernet connection when programmed in accordance to IEC-60488 Protocol (former IEEE-488 Protocol).

3.9 Triggering Voltages. The electronic calibrator triggering voltages shall be compliant with the IEC-60488 Protocol (former IEEE-488 Protocol).

3.10 Input Impedance. The electronic calibrator input impedance shall be ≥10 Mega ohms in DC and AC operation throughout the bandwidth.

3.11 Design and Construction. The electronic calibrator shall be used in both bench top and rack mounted environments. The electronic calibrator is a commercial off the shelf item subject to commercial products regulatory standards. The electronic calibrator shall comply with all applicable MIL-PRF-28800 sections. The electronic calibrator shall be designed and constructed in accordance with all applicable European Union (EU) requirements in order to have the "CE" marking affixed.

3.12 System Safety. The system shall comply with all sections of the Underwriters Laboratories (UL), IEC/ISO standard 61010A and 61010B for electrical safety.

3.13 Sound Levels. The electronic calibrator acoustic noise shall conform to sound level requirements on section 1 of ISO 7779 and sections 1 & 2 of ISO 8253.

3.14 Electrostatic Discharge (ESD). The design of the electronic calibrator shall preclude equipment damage due to ESD (internal/external electrical arcing), protect personnel from electrical shock due to static charging, and prevent ignition of explosive atmospheres due to sparking in accordance with MIL-STD-1686 sections 5.2 and 5.7.2 and MIL-HDBK-263 for guidance.

3.15 Electro Magnetic Interference and Compatibility (EMI/EMC). The electronic calibrator EMI and EMC emissions shall be in accordance with the following radiated emission and susceptibility requirements of MIL-PRF-28800 section 4.5.6.5, MIL-STD-461: RE102 and RS103 & IEC-61000-4 thru IEC-61000-6. Emitted radiation absorption and radiation limits FCC Part 15.109 & 15.209.

3.16 Port Damage. The input ports shall withstand up to 76 dBV or approximately 1,500 Volts (typical) levels without damaging the internal circuitry, protection can be achieved by either internal or external fusing.

3.17 Port Overvoltage Protection. The electronic calibrator shall have an audible, visual or both methods of warning the user in the event that a voltage exceeding the nominal rating is applied to the input port.

3.18 Connectors. The system shall have "Banana" type connectors and BNC & RS-232 type connectors shall be available for other rear panel inputs and outputs functions. The system shall



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have a General Purpose Interface Bus (GPIB) IEC-60488-2 (IEEE-488.2) connector required for compatibility and control functions.

3.19 Operating Temperature Range. The electronic calibrator shall meet the specifications listed herein with full capabilities within the temperature range of 32° to 122°F (0°C to 50°C) after performing tests in MIL-PRF-28800 section 4.5.5.1.1 table 11 for class 3 equipment.

3.20 Storage Temperature Range. The electronic calibrator shall meet and/or exceed manufacturer's specifications for stored equipment after tests of MIL-PRF-28800 section 4.5.5.1.1 table 11 for temperature range and relative humidity storage for class 3 equipment classification.

3.21 Altitude Requirements. The electronic calibrator shall meet its operating performance and accuracy while operating at an altitude equal or greater than 5,000 ft. (1,524 meters). The equipment shall meet performance and accuracy requirements, for non-operating storage up to 10,000 (3,050 meters).

3.22 Calibration & Maintenance Adjustments. No special tools or software shall be required for alignment or adjustment except those furnished with the electronic calibrator test set.

3.23 Calibration Period. The design of the electronic calibrator shall provide readily accessible calibration and maintenance adjustments. The calibration adjustments shall be accessible without removal of the instrument internal modules. The calibration interval shall be based on a period of one calendar year or an operating time of 2,080 hours or whichever occurs first.

3.24 Reliability. The design of the electronic calibrator shall be such that under normal use and operation the mean time between failures (MTBF) exceeds the minimum of 1,500 hours of operation accordance with MIL-PRF-28800. The manufacturer shall provide a summary of maintenance or warranty records to establish the validity of the statistical assertion in accordance with MIL-PRF-28800 section. The manufacturer shall be able to reasonably establish the claim that the offered equipment is capable of the required level of reliability by meeting and/or exceeding MIL-PRF-28800 sections and section.

3.25 Maintainability The electronic calibrator shall meet the maintainability criterion in accordance with MIL-STD-1472 paragraphs 5.9 through 5.9.18 and 5.9.21 and MIL-PRF-28800 sections 3.14.1 through 3.14.4 and sections 6.6.2.4 and 6.6.24.1.

3.26 Preventive Maintenance. Preventive maintenance shall not require more than 15 minutes per 30-day period. Preventive maintenance shall not require breaking of the equipment seams where calibration seals would normally be placed in accordance with MIL-PRF-28800 sections 3.14.2 & 3.14.3.

3.27 Corrective Maintenance. The electronic calibrator shall be designed to meet reparability criterion in accordance with section 3.14.3 sub sections a through d of MIL-PRF-28800.



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3.28 Electrical Power Sources & Connections. The electronic calibrator shall operate from nominal commercial, military, aircraft and power sources with a nominal voltage of 120 $V_{RMS}$ Single Phase AC (±10%) at line frequencies of 50 Hz & 60 Hz (±5%), in accordance with MIL-PRF-28800 paragraph 3.1.5.2 and 3.1.5.2a. Transient state conditions shall meet MIL-PRF-28800 requirements for the classification specified herein. AC input power shall be used to power the SA and recharge batteries for nonvolatile memory if equipped with them.

3.29 Dimensions. The electronic calibrator shall comply with width and depth of EIA/ECA-310 standard equipment rack configuration. The width shall be the 19” when configured with rack adapter/mount hardware and include a rack-mount conversion kit if required at organizational level. The height shall be compatible with standard EIA height units. One unit or "U" is defined by EIA/ECA-310 to be 1.75 inches or 44.45 mm.

3.30 Weight. The electronic calibrator system shall comply with MIL-STD-1472 section 5.9.11.3.3 for weight & size restrictions for single person lift & handling. The total weight of the electronic calibrator, excluding accessories and manuals, shall not exceed a two-person lift of 70 pounds, (32 kilograms).

## 4. REGULATORY REQUIREMENTS.

4.1 Recycled recovered materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or refurbished items shall not be provided.

4.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

## 5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and the product conforms to the producer’s engineering drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such



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conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

5.2. Verification. Verification of the delivered Electronic Calibrator will be performed at the delivery point by the end user.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Source of documents.

7.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained online at http://assist.dla.mil/quicksearch or from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

7.1.2 IEC documents may be obtained online at http://www.iec.ch/ or from IEC Regional Centre for North American (IEC-ReCNA) 446 Main Street, 16th Floor, Worcester, MA 01608.

7.1.3 FAR documents may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of the FAR may be obtained online from https://www.acquisition.gov/far/ .

7.1.4 EIA/ECA documents may be obtained online at http://www.global.ihs.com/ or from IHS Customer Care 15 Inverness Way East, Englewood CO. 80112.

7.1.5 FCC documents may be obtained online at http://www.ecfr.gov/ or from the Federal Communications Commission 445 12th Street SW, Washington DC 20554.

7.1.6 ISO documents may be obtained online at http://www.iso.org/ or from the American National Standards Institute 1899 L Street NW, 11th Floor US-Washington, DC 20036.

7.2 National Stock Number (NSN). The following NSN corresponds to this CID. The NSN may not be indicative of all possible NSNs associated with this CID. (For reference only)

NSN: 6625-01-520-5111



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7.3 Key words.

Alternating Electricity
Alternating Potential
Direct Electricity
Direct Potential
Electronic Calibrator
Resistance Values

Custodians:
Air Force – 84

Reviewers:
Air Force – 99

Preparing Activity:
Air Force – 84

Agent:
Air Force – 99

(Project 6625-2013-043)

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